03 OSI vs TCP IP Model

1. What Is a Network Model?

A **network model** is a structured framework that explains how computers communicate and exchange data over a network.

It divides network communication into different layers. Each layer performs a specific function and works with other layers to deliver data from one device to another.

Why Do We Need Network Models?

- To understand how data travels between computers.
- To make network design easier.
- To troubleshoot network problems.
- To standardize communication between different devices.
- To understand where network security problems may occur.

Types of Network Models

There are two important network models:

1. **OSI Model:** Has 7 layers.
2. **TCP/IP Model:** Commonly explained using 4 layers.

2. OSI Model (Open Systems Interconnection)

The **OSI model** is a reference model that explains network communication through seven layers.

Each layer has a specific responsibility.

The Seven OSI Layers

Layer No.	Layer Name	Main Function	Examples
7	Application	Provides network services to applications	HTTP, DNS, SMTP
6	Presentation	Handles data format, encryption and compression	Data encoding, TLS-related presentation functions
5	Session	Establishes, manages and ends communication sessions	Session management, RPC concepts
4	Transport	Provides end-to-end data delivery and port addressing	TCP, UDP
3	Network	Handles logical addressing and routing between networks	IP, ICMP
2	Data Link	Transfers frames within a local network using MAC addresses	Ethernet, Wi-Fi MAC, ARP
1	Physical	Transmits raw bits through physical or radio signals	Cables, connectors, radio signals

Important: TLS is commonly associated with the application security stack, although it provides functions often used to explain the OSI Presentation layer. ARP sits between Layer 2 and Layer 3 in practical networking and is commonly associated with the link layer.

Layer 7: Application Layer

Function: Provides network services used by applications.

Examples:

- HTTP and HTTPS for websites.
- DNS for resolving domain names.
- SMTP for sending email.
- FTP for file transfers.

Security risk: Application-layer attacks such as SQL injection or malicious HTTP requests.

Example: An attacker sends a specially crafted request to exploit a vulnerable website.

Layer 6: Presentation Layer

Function: Represents data in a format that the receiving system can understand. It is commonly associated with encoding, encryption and compression concepts.

Examples:

- Character encoding such as UTF-8.
- Data formats such as JSON.
- Encryption and decryption concepts.

Security risk: Weak or incorrectly implemented encryption and insecure data handling.

Example: Sensitive information is exposed because an application transmits it without adequate encryption.

Layer 5: Session Layer

Function: Establishes, manages and terminates communication sessions between applications.

Examples:

- Session management.
- Remote Procedure Call (RPC) concepts.
- Application session handling.

Security risk: Session hijacking.

Example: An attacker steals a valid session token and uses it to impersonate a logged-in user.

Layer 4: Transport Layer

Function: Supports end-to-end communication, segmentation, reliability and port-based delivery.

Main protocols:

- **TCP:** Connection-oriented and reliable.
- **UDP:** Connectionless with lower overhead.

Security risk: TCP SYN flood or UDP flood attacks.

Example: An attacker sends excessive connection requests, consuming server resources.

Layer 3: Network Layer

Function: Provides logical addressing and routes packets between different networks.

Main protocols:

- IPv4 and IPv6.
- ICMP.
- Routing protocols such as OSPF.

Security risk: IP spoofing or certain routing attacks.

Example: An attacker forges a source IP address to make traffic appear to come from another device.

Layer 2: Data Link Layer

Function: Transfers frames within a local network and uses MAC addressing for local delivery.

Examples:

- Ethernet.
- Wi-Fi MAC functions.
- VLANs.
- ARP, which is used to resolve IPv4 addresses to MAC addresses on local networks.

Security risk: ARP spoofing or MAC flooding.

Example: An attacker on an inadequately protected local network sends false ARP messages to redirect traffic.

Layer 1: Physical Layer

Function: Transmits raw bits using electrical, optical or radio signals.

Examples:

- Ethernet cables.
- Fiber-optic cables.
- Network connectors.
- Radio signals used by Wi-Fi.

Security risk: Physical cable tampering or unauthorized access to networking equipment.

Example: An unauthorized person connects a device to an exposed network port.

3. Easy Trick to Remember the OSI Layers

From Layer 7 to Layer 1:

A P S T N D P

- **A** — Application
- **P** — Presentation
- **S** — Session
- **T** — Transport
- **N** — Network
- **D** — Data Link
- **P** — Physical

Memory sentence:

“All People Seem To Need Data Processing.”

Remember the order from top to bottom as Application → Physical.

4. TCP/IP Model

The **TCP/IP model** explains how real-world internet communication works. It is commonly divided into four layers.

Layer	Main Function	Example Protocols
4. Application	Provides application-level network services	HTTP, HTTPS, DNS, SMTP, SSH, DHCP
3. Transport	Provides process-to-process communication	TCP, UDP
2. Internet	Handles IP addressing and packet routing	IPv4, IPv6, ICMP
1. Network Access	Handles local network delivery and physical transmission	Ethernet, Wi-Fi

Some textbooks use the term **Network Interface** instead of Network Access.

TCP/IP Application Layer

Combines the responsibilities associated with the OSI Application, Presentation and Session layers.

Examples:

- HTTP/HTTPS.
- DNS.
- SMTP.
- SSH.

TCP/IP Transport Layer

Provides end-to-end communication using TCP or UDP.

Examples:

- TCP for reliable communication.
- UDP for low-overhead communication.

TCP/IP Internet Layer

Routes packets between networks using logical IP addresses.

Examples:

- IPv4.
- IPv6.
- ICMP.

TCP/IP Network Access Layer

Handles communication over the local network and the physical transmission of data.

Examples:

- Ethernet.
- Wi-Fi.

5. OSI vs TCP/IP Model

Basis	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Number of layers	7	Commonly 4
Type	Reference model	Protocol suite and layered model
Purpose	Helps explain network communication	Supports practical internet communication
Application functions	Three separate upper layers	One Application layer
Lower layers	Data Link and Physical are separate	Combined as Network Access
Use	Learning, troubleshooting and design	Internet and network implementation
Examples	A conceptual framework	TCP, IP, HTTP, DNS

Mapping OSI to TCP/IP

OSI Layers	Corresponding TCP/IP Layer
Application (7)	Application
Presentation (6)	Application
Session (5)	Application
Transport (4)	Transport
Network (3)	Internet
Data Link (2)	Network Access
Physical (1)	Network Access

Key point: The OSI model has seven layers, while the commonly taught TCP/IP model has four layers.

6. Protocols and Examples by Layer

OSI Layer	Protocols / Technologies	Example
Application	HTTP, HTTPS, DNS, SMTP, FTP	Opening a website
Presentation	Data encoding, encryption and compression	Representing data in UTF-8
Session	RPC and session management concepts	Managing an application session
Transport	TCP, UDP	Delivering data to the correct port
Network	IPv4, IPv6, ICMP	Routing a packet to another network
Data Link	Ethernet, Wi-Fi MAC, ARP	Delivering a frame on a local network
Physical	Ethernet physical standards, fiber, radio	Transmitting bits over a cable

Note: The OSI model is a conceptual framework, so not every real-world protocol maps perfectly to one specific layer.

7. Security Risks and Attacks by OSI Layer

OSI Layer	Example Attack or Risk	Basic Protection
7. Application	SQL injection, cross-site scripting (XSS)	Secure coding, input validation, WAF
6. Presentation	Weak encryption or insecure data encoding	Strong encryption and secure data handling
5. Session	Session hijacking	Secure cookies, session rotation, HTTPS
4. Transport	TCP SYN flood	Firewalls, rate limiting, SYN flood protection
3. Network	IP spoofing	Ingress/egress filtering and network controls
2. Data Link	ARP spoofing	Dynamic ARP Inspection and network segmentation
1. Physical	Cable tampering or unauthorized device connection	Physical access controls and port security

These are representative examples. Many attacks span multiple layers, and the OSI model does not mean that every attack belongs exclusively to one layer.

8. How Data Travels Through the Layers

Suppose you open https://example.com in a browser.

1. **Application:** The browser prepares an HTTP request and uses DNS when it needs the server's IP address.
2. **Presentation:** Data formatting and encryption functions are applied where appropriate, such as TLS encryption for HTTPS.
3. **Session:** Communication sessions are managed by the relevant application and protocol mechanisms.
4. **Transport:** TCP carries the data in a typical HTTPS-over-TCP connection. HTTP/3 instead uses QUIC over UDP.
5. **Network:** IP addresses are used to route packets toward the destination.
6. **Data Link:** Ethernet or Wi-Fi carries frames across the local link.
7. **Physical:** Bits travel through cables, fiber or radio signals.

At the receiving device, the data is processed upward through the relevant protocol layers.

Simple Diagram

Your Computer → Application → Transport → Internet/Network → Network Access → Network

The receiving server processes the incoming data through its own protocol stack.

9. Important Terms

- **Packet:** A unit of data handled by the network layer.
- **Frame:** A unit of data handled by the data link layer.
- **Segment:** A unit of TCP data; UDP uses the term datagram.
- **Protocol:** A set of rules for communication.
- **IP address:** A logical address used to identify and route traffic to a network interface.
- **MAC address:** A link-layer address used for local network delivery.
- **Port:** A number used to identify a network service or application endpoint.
- **Encapsulation:** The process of adding protocol headers as data moves down the network stack.

10. Quick Revision

Question	Short Answer
How many OSI layers are there?	7
How many TCP/IP layers are commonly taught?	4
Which OSI layer handles routing?	Network
Which layer uses TCP and UDP?	Transport
Which layer provides services to applications?	Application
Which layer handles MAC addressing?	Data Link
Which layer transmits raw bits?	Physical
What is encapsulation?	Adding protocol information as data moves down the stack
What is a common Layer 7 attack?	SQL injection
What is a common Layer 2 attack?	ARP spoofing

Conclusion

The OSI and TCP/IP models help us understand how data travels between devices. The OSI model divides communication into seven layers, while the commonly taught TCP/IP model uses four layers. Understanding their functions, protocols and security risks helps cybersecurity professionals troubleshoot networks, investigate incidents and identify where security controls are needed.